Wages and Employment in a Single Labour Market

EC306- Wages and Employment

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Fall 2025

Goals of This Section

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- Bring together demand and supply to determine wages and employment
- Discuss the effect of market power in goods market on equilibrium wages and employment
- Discuss the effect of market power in the labour market on equilibrium wages and employment
- Analyze minimum wages

Introduction

Introduction

- We have analyzed labour demand and supply separately in detail
- The equilibrium wage and employment is determined by the interaction of the two
- Market power plays an important role
  - In both goods markets and labour markets
- With a full understanding on how wages and employment are determined, we can analyze policies
- The minimum wage is one of the most prominent labour market policies
  - In theory, it leads to unemployment
  - The evidence is more mixed

Competitive Firms

Introduction

- We start with firms that are competitive in the goods and labour markets
- Perfect competition in both implies that
  - Firms take the market wage as given
  - They take the market price as given
  - Competition means they are too small to influence either
- We analyze the equilibrium wage and employment for a homogeneous type of labour
  - All workers are the same

Market Demand and Supply

- The market demand and supply are the sum of the individual firms' demand and supply at each wage
- Here we determine the demand and supply curve for a single labour market
  - Could be an specific occupation in a specific region
  - Example: electricians in Ontario, professors in Canada, etc.
  - How the market is defined depends on context
  - But it is for a single type of labour

Market Demand and Supply

- Example is two firms with different elasticities of demand
  - Each firm demand curve is their value of marginal product
  - Firms may face different product prices if they sell different things (but hire the same labour)
- Each firm faces the same wage in the labour market
  - They perceive the supply curve as horizontal in the long run
  - They take the wage as given and can get as much laour as they want at that wage
  - Market supply is upward sloping
- Market demand is horizontal sum of individual demands
  - Add up the quantities demanded at each wage for each firm
  - Repeat for all wages

Market Demand and Supply

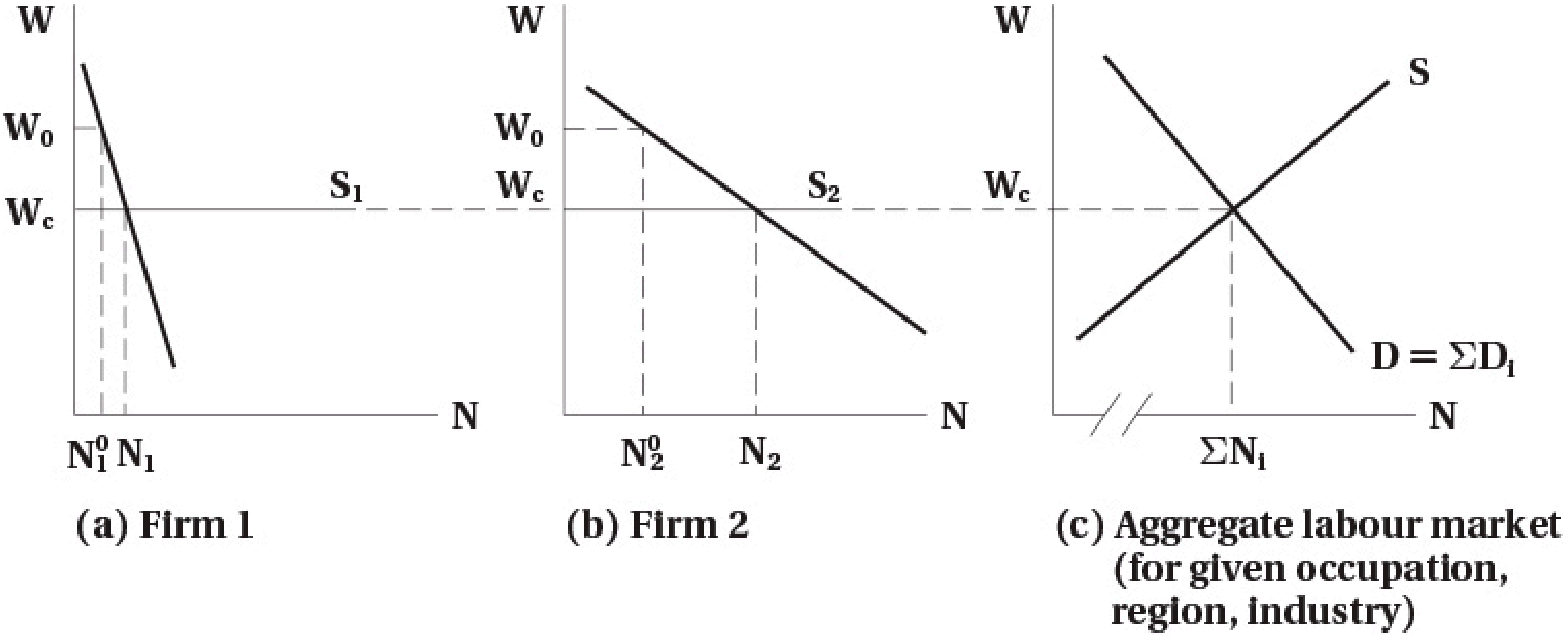


FIGURE 7.1 Competitive Product and Labour Markets

Firm Demand and Supply in the Short Run

- Above we analyzed the longer run
  - Firms can get as much labour as they want at the going wage
- In the short run, firms may face an upward sloping supply curve
  - They have to pay higher wages to attract more workers
  - This could be because of contracts, or other frictions
- If demand rises, they would pay higher wages to attract workers
- But because wages are high, over the long term more workers enter the market
  - Short run supply shifts out
  - Wage falls back down to long run level

Firm Demand and Supply in the Short Run

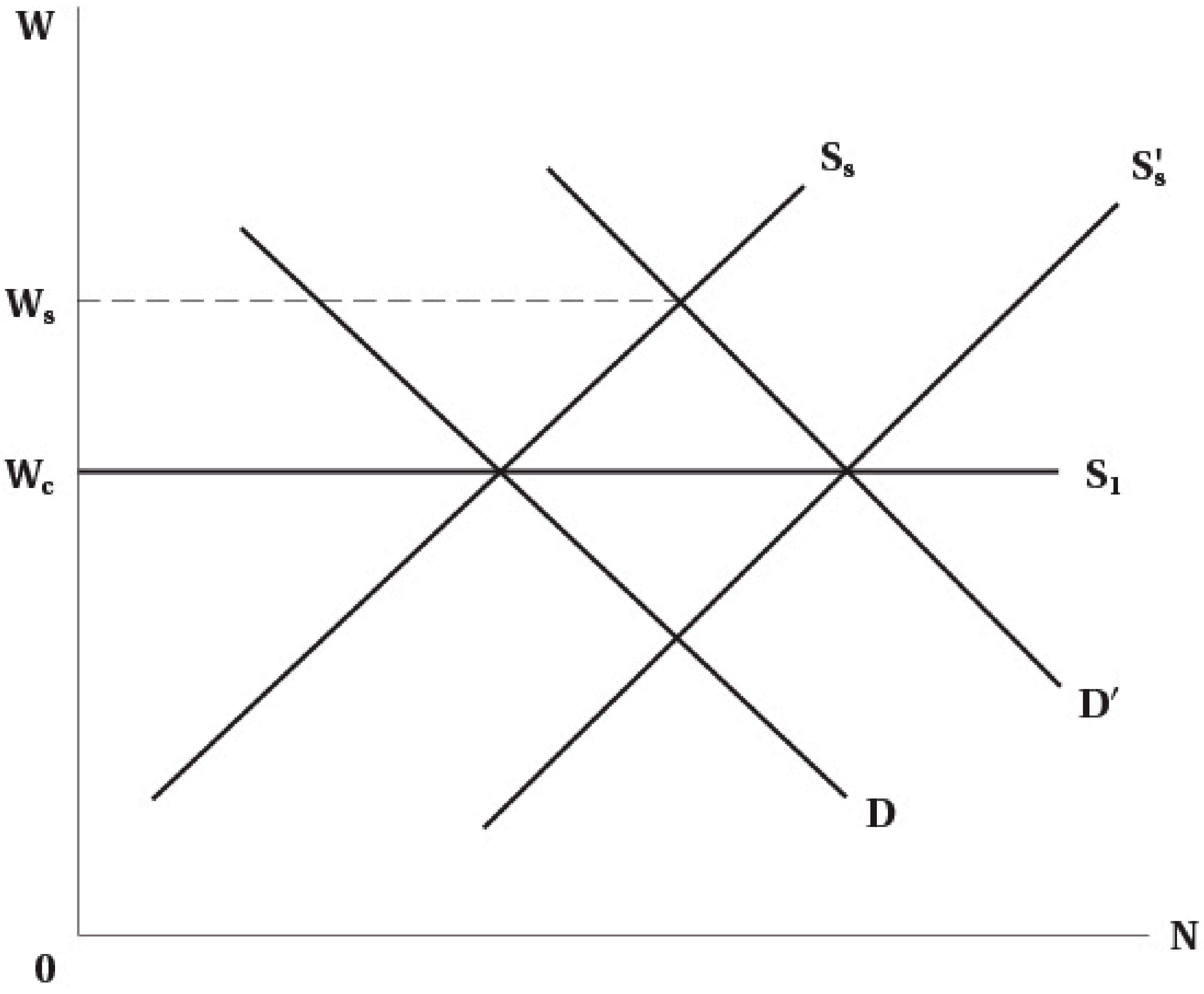


FIGURE 7.2 Short-Run and Long-Run Labour Supply Schedules to the Firm

Imperfect Competition in the Product Market

Monopoly

- In competitive markets there are many sellers of a product
- In a monopoly, there is only one
- This gives the firm market power
  - The ability to set prices above marginal cost
- For a monopolist, the demand curve is where wage equals marginal revenue product

$$w = MR \times MPP_N$$

- For a monopolist, marginal revenue is less than price
  - Because to sell more, they have to lower price on all units sold
  - So marginal revenue product is less than value of marginal product (VMP)

Monopoly

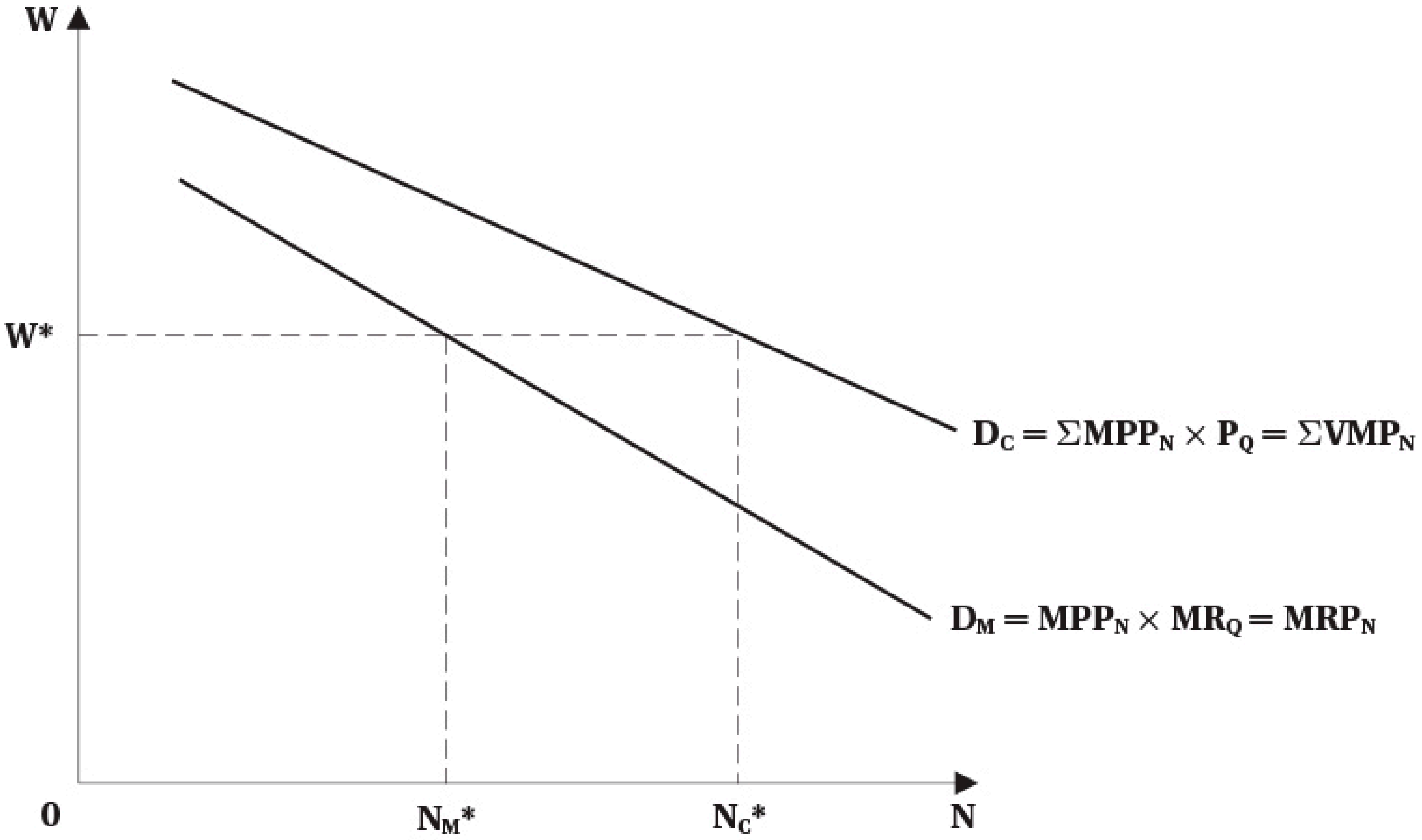


FIGURE 7.3 Monopolist versus Competitive Demand for Labour

Monopoly

- Because monopolists have market power
  - They can raise the price of their product above competitive level
  - This will decrease output
  - They demand less labour
- Monopolists therefore will hire fewer workers
  - Compared to a competitive firm in the same market
- Note that power in the product market does not mean power in the labour market
  - The firm is still one of many hiring workers in a particular labour market

Monopolistic Competition and Oligopoly

- Many markets are neither perfectly competitive nor pure monopoly
- Monopolistic competition
  - Many firms selling differentiated products
  - Each firm has some market power because of product differentiation
  - Can set prices above marginal cost
- Oligopoly
  - A few firms dominate the market
  - Each firm has significant market power
  - But must consider reactions of competitors when setting prices
- Demand for labour is still based on marginal revenue product
  - Which is less than value of marginal product
  - So firms hire fewer workers than in competitive markets

Additional Details

- All of the previous analysis assumes that labour markets are perfectly competitive
  - Firms can hire all labour they want at the given wage
- That is not always the case in reality when firms have power in the product market
  - Employees may unionize
  - Firms may hire a large number of the market's workers
- In these cases, firms may not face a perfectly elastic supply curve for labour
- Later we will analyze imperfect competition in the labour market

Math of Labour Market Equilibrium

Introduction

- We know that market equilibrium is where demand and supply are equal
- You can do that graphically or mathematically
- In this section we cover both
- To make the math easier, we use linear demand and supply curves
  - This is just for simplicity
  - The same principles apply with non-linear curves

Demand and Supply Functions

- The demand and supply for labour are functions of wages and other factors
  - Demand: $N^D = f(w, Z)$
    - Z is a list of other factors affecting demand
  - Supply: $N^S = g(w, X)$
    - X is a list of other factors affecting supply
- Z and X are things that shift the curves
  - For example, technology, prices of other inputs, preferences, demographics, etc.
  - We call these **exogenous** variables
    - Determined outside the model
- N^D, N^S, and w are **endogenous**
  - Determined within the model

Demand and Supply Functions

- Assume the demand and supply functions are linear
  - Demand: $N^D = a + bw$
    - a, b are constants
  - Supply: $N^S = c + fw$
    - c, f are constants
- To find equilibrium, set demand equal to supply

$$a + bw^* = c + fw^*$$

Demand and Supply Functions

- Rearranging to solve for equilibrium wage:

$$w^* = \frac{c - a}{b - f}$$

- Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af}{b - f}$$

- Use these equations to analyze how changes in exogenous variables
  - Changing a would shift the demand curve
    - Example: increase in product price
  - Changing c would shift the supply curve
    - Example: increase in population

Demand and Supply Functions

- To draw the demand and supply curves, we need w on the left side
- Rearranging the demand function:

$$w = \frac{N^D - a}{b}$$

- Rearranging the supply function:

$$w = \frac{N^S - c}{f}$$

- Next slide plots these functions

Demand and Supply Functions

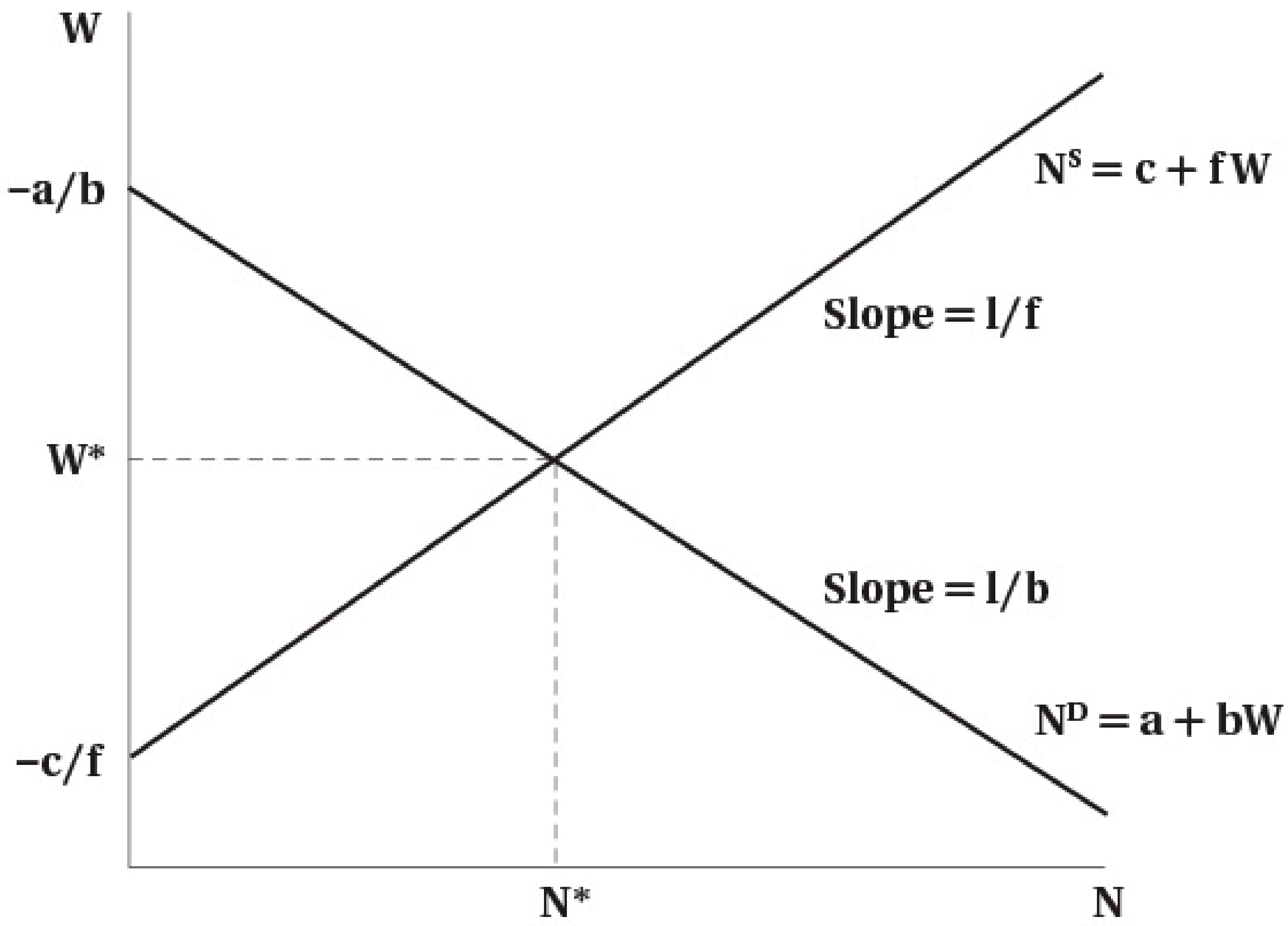


FIGURE 7.4 Linear Supply and Demand Functions

Payroll Tax

- We can use this model to analyze various policies
- One common policy is a payroll tax on employers
  - Examples: Canada Pension Plan (CPP), Employment Insurance (EI)
- These are typically a percentage of wages paid by employers
- But we will analyze a fixed amount per hour for simplicity
  - Suppose the tax is T per hour
- This drives a "wedge" between the wage paid by employers and the wage received by workers in equilibrium

$$w_r = w_e - T$$

Payroll Tax

- The demand function is:

$$w_e = \frac{N^D - a}{b}$$

- The supply function is:

$$w_r = \frac{N^S - c}{f}$$

Payroll Tax

- Substitute into the equilibrium price equation and set quantity demanded equal to quantity supplied:

$$\frac{N^* - c}{f} = \frac{N^* - a}{b} - T$$

- Rearranging to solve for equilibrium employment:

$$N^* = \frac{bc - af - bfT}{b - f}$$

Payroll Tax

- The equilibrium wage received by workers is

$$w_r^* = \frac{c-a}{b-f} - \frac{b}{b-f}T$$

- The equilibrium wage paid by employers is

$$w_e^* = \frac{c-a}{b-f} - \frac{f}{b-f}T$$

Payroll Tax

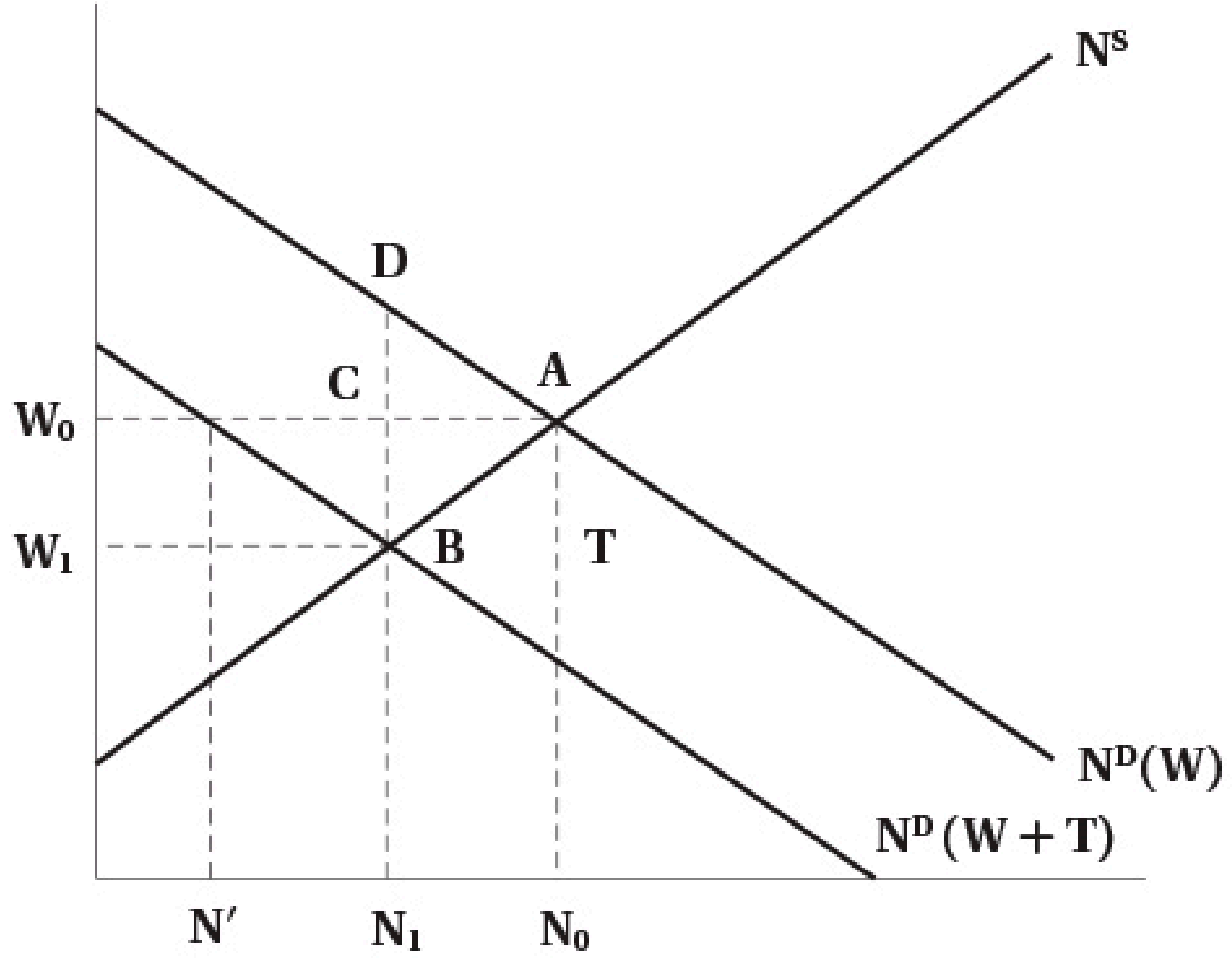


FIGURE 7.5 The Effect of a Payroll Tax on Employment and Wages

Payroll Tax

- The previous slide highlights the difference between the statutory and economic incdidence of a tax
  - Statutory incidence: who is legally responsible for paying the tax
    - In this case, firms
  - Economic incidence: who actually bears the burden of the tax
    - Shared between firms and workers
    - Workers receive a lower wage than before
    - Firms pay a higher one
- Economic incidence depends on elasticities of supply and demand
  - More elastic supply means workers bear less of the tax burden
  - More elastic demand means firms bear less of the tax burden

Evidence

- Theory predicts lower employment from a payroll tax
- Also predicts economic incidence may be shared between firms and workers
- Evidence based on empirical studies is
  - In the long run, workers bear most of the burden of the tax through lower wages
  - Employment effects are small

Monopsony in the Labour Market

Introduction

- Market power can be a feature on both the buying and selling side of markets
  - Monopoly: single seller with market power
  - Monopsony: single buyer with market power
- Typically we discuss monopoly in the goods market, but monopsony in the labour market
  - But you can have power in both buying and selling in both markets
- In this section we will consider monopsony in the labour market

Introduction

- Some examples of monopsony in the labour market
  - Company/factory towns
  - Hospital labour markets
  - University professor labour markets
  - Public sector in small areas
- In all case there is a large buyer of labour
- This gives the buyer power to set wages
- Affects both wages and employment
  - To attract labour, it must raise wages

Simple Monopsony

- In a monopsony, the firm faces the market supply curve for labour
  - Supply is upward sloping
  - Contrast to perfect competition where supply is horizontal
  - To hire more workers, the firm must pay a higher wage
- The firms marginal cost of labour is above the supply curve
  - Because to hire an additional worker, the firm must raise the wage for all workers
  - So MC is equal to cost of additional worker plus the increase in wage paid to existing workers
  - Analogous to how monopolists face MR below demand

Simple Monopsony

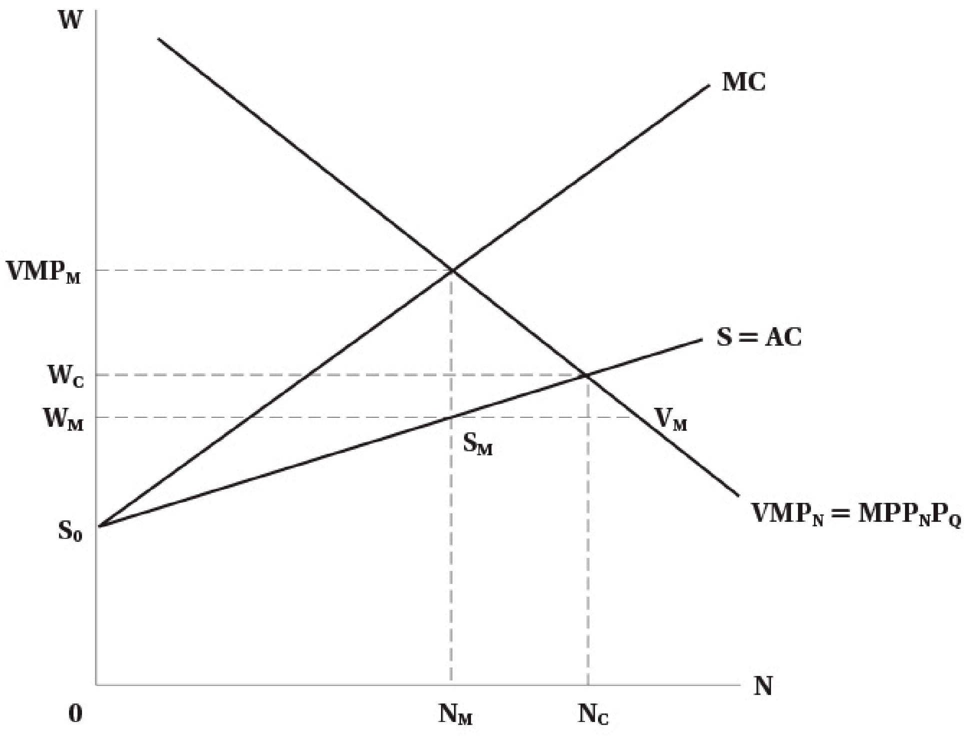


FIGURE 7.6 Monopsony

- Monopsony depicted to the left
- Firm demand is its value of marginal product
- Faces upward sloping supply curve
- MC is above supply
- Optimal labour is where MC = VMP
- Wage comes from supply curve at that quantity of labour

Simple Monopsony

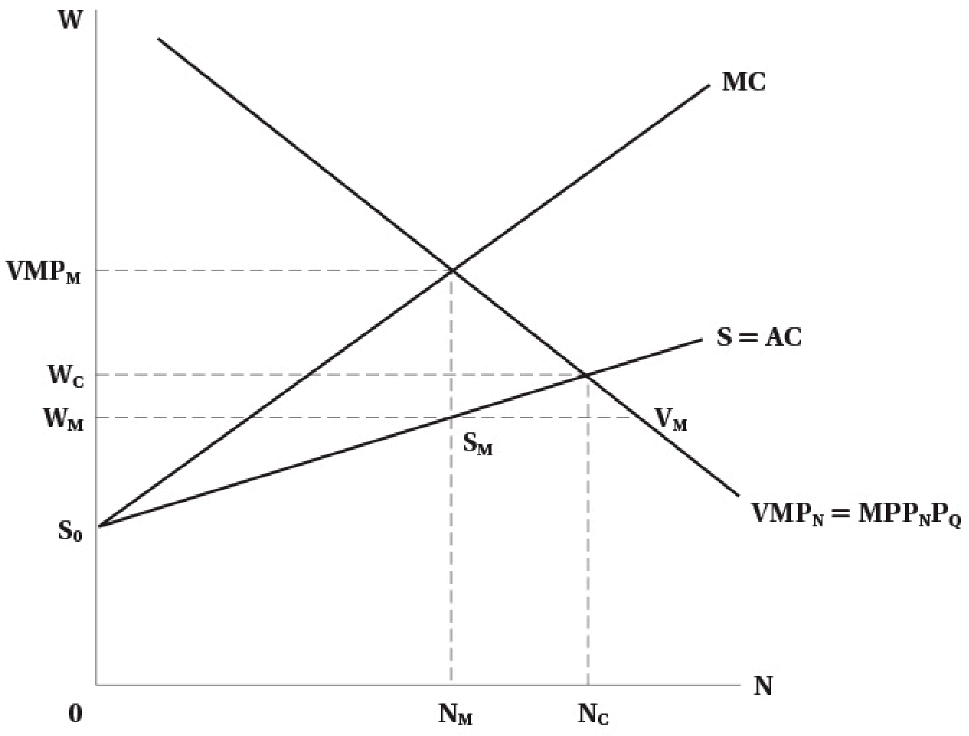


FIGURE 7.6 Monopsony

- For monopsony
  - Wage is below VMP
  - Wage is below competitive wage
  - Employment is below competitive employment
- Monopsony profit is rectangle $(VMP_m - W_m) \times N_m$
  - Wage bill is $W_m \times N_m$
  - Revenue contributed by workers is $VMP_m \times N_m$

Simple Monopsony

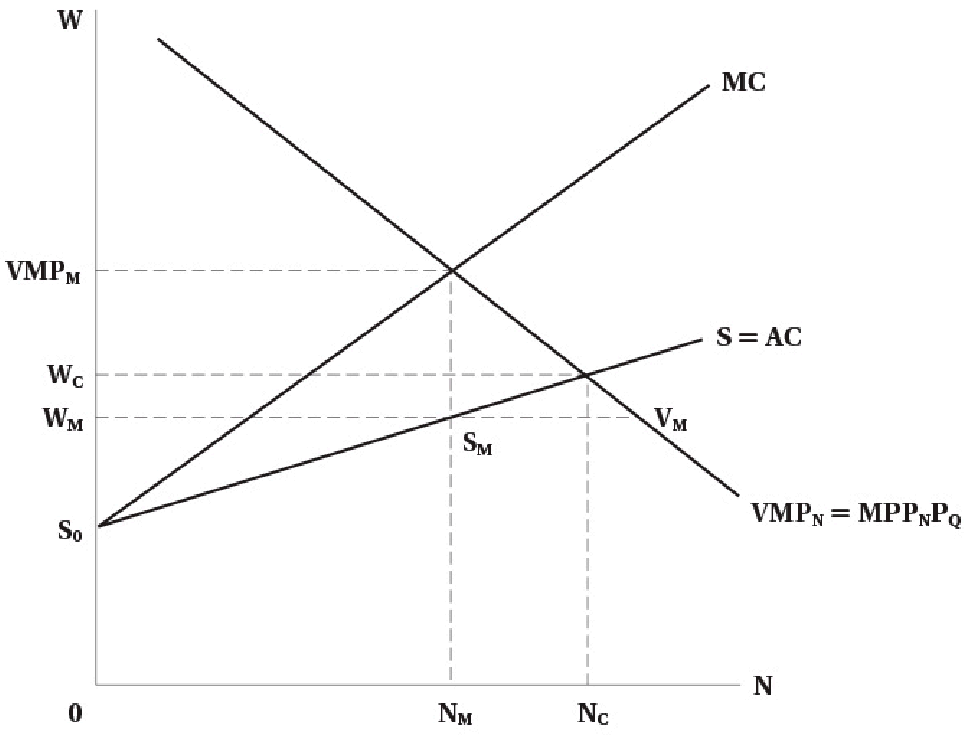


FIGURE 7.6 Monopsony

- Monopsonies also leave jobs vacant
  - Because they restrict employment to raise wages
  - This creates inefficiency in the labour market
- Vacancies are $V_m - S_m$
- At wage W_m
  - VMP is above wage
  - Firms would like to hire more -But to hire it must raise wages, so it is optimal to not hire more

Characteristics of Monopsonies

- More prevalent in the short run
  - In the long run, workers can move to find other jobs
  - In the short run, frictions may prevent this
- Large relative to the labour market
  - Does not have to be a big firm, just large relative to the market
  - If the market is small, the firm does not have to be large to have power
  - Example: a restaurant in a seasonal tourist town
- Workers have specialized skills
  - May naturally be few employers for specialized labour
  - Like academia, sports teams

Characteristics of Monopsonies

- Wages are lower than competitive equilibrium, but not necessarily low in an absolute sense
  - Example: sports players
  - Not paid low wages, but market power makes them lower than they would be otherwise

Wage Differentiation

- In previous analysis, we assumed all workers are the same
  - Single supply curve for labour
- This leaves some surplus for workers
  - There are workers willing to work for less than the equilibrium wage
- Firm can capture that surplus by differentiating wages
  - Paying different workers different wages based on their reservation wage

Wage Differentiation

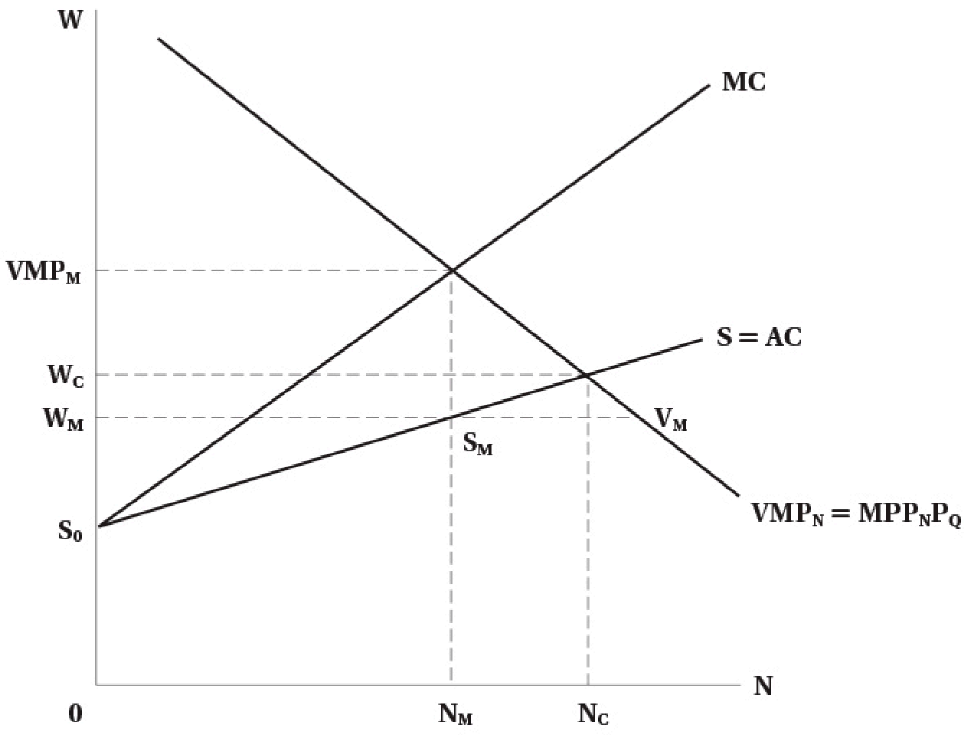


FIGURE 7.6 Monopsony

- If the firm can pay different wages
  - It can pay each worker their reservation wage
  - MC curve and supply curve become equal to supply curve
- Firm will hire up to competitive level
  - Because it does not have to increase wage for all workers to hire more
- To do this, firms would try to make wages secret
  - Prevent workers from knowing what others are paid
  - Could do this with non-wage benefits for specific workers

Evidence of Monopsonies

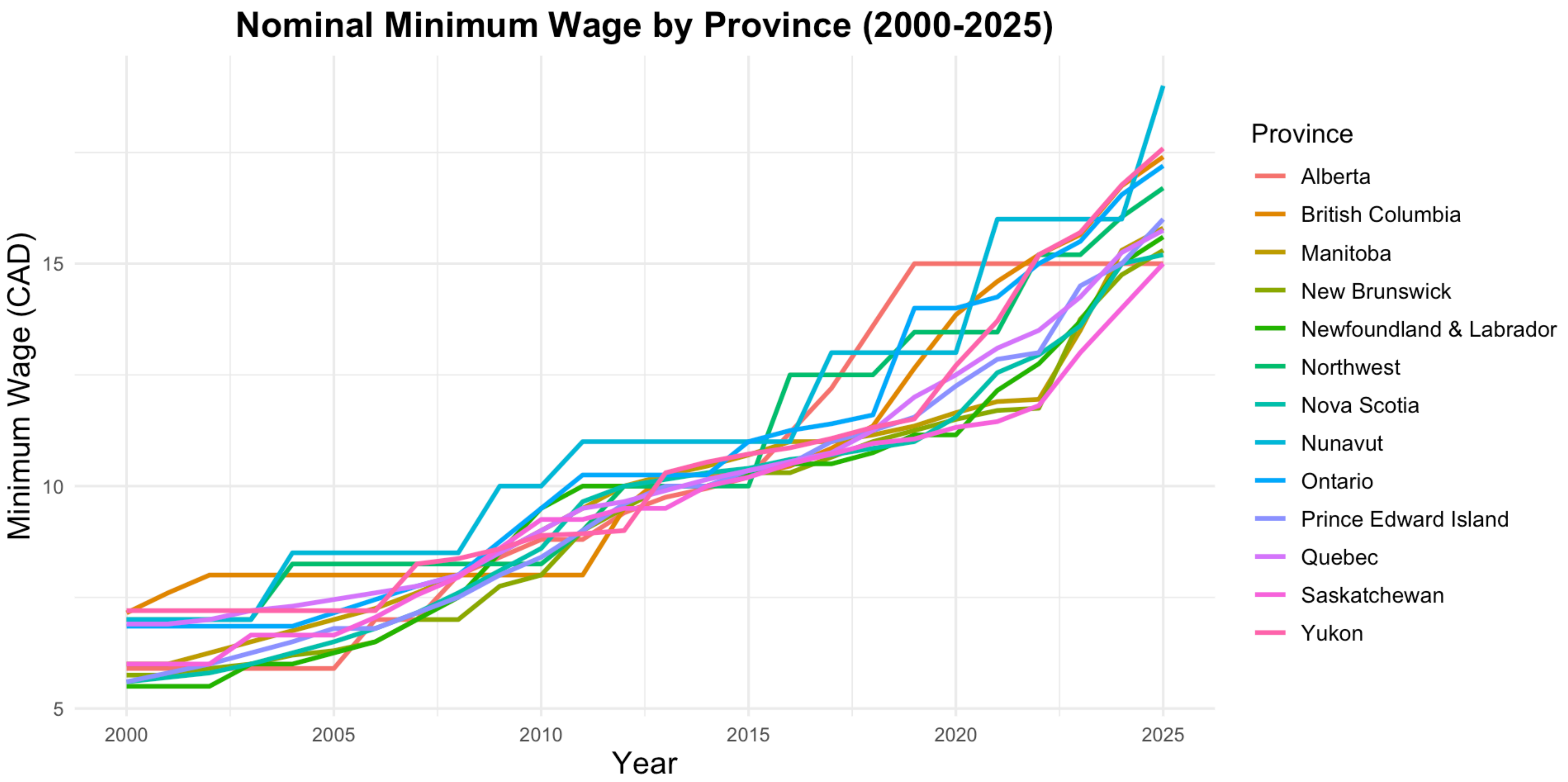
- Pro sports
  - Baseball: free agency increased wages significantly
    - Increases competition among teams
  - Hockey: not much evidence of monopsony power
- Academia and nursing
  - Some evidence of monopsony power
  - In Canada, a bargaining model may be more appropriate given these labour markets are unionized
- Some academics argue that monopsonies are relatively common
  - Firms can set wages only if they have some market power
  - Evidence that this is widespread

Minimum Wages

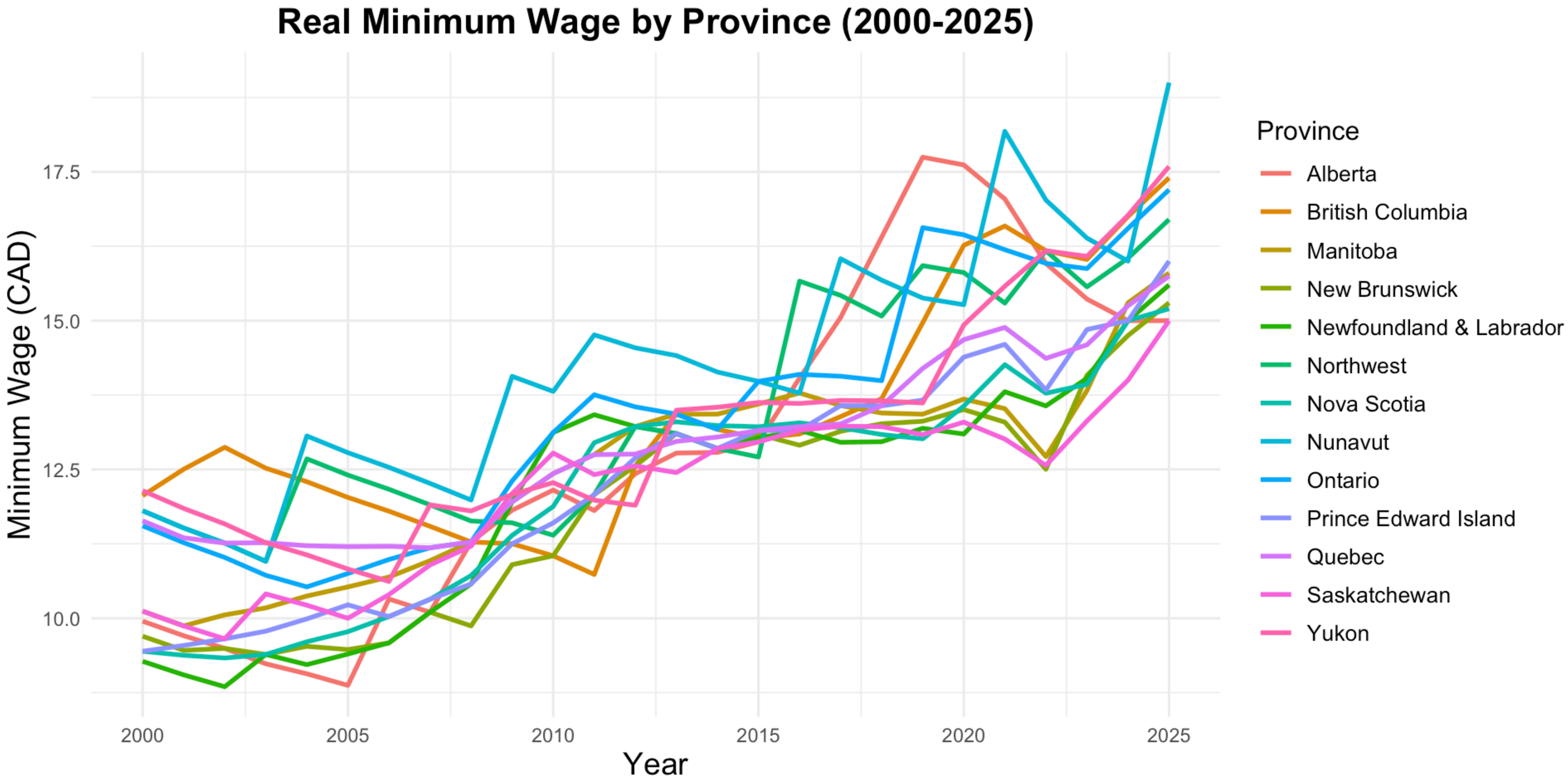
Introduction

- Minimum wages are a common policy tool
  - Set a legal minimum wage that employers must pay workers
- Intended to raise wages of low-wage workers
  - Reduce poverty
  - Increase standard of living
- Standard economic theory predicts that minimum wages lead to unemployment
  - Because they set a wage above the equilibrium wage
  - Quantity of labour supplied exceeds quantity demanded
- But statistical evidence is much more mixed

Minimum Wages by Province



Minimum Wages by Province



Minimum Wages in Canada and USA

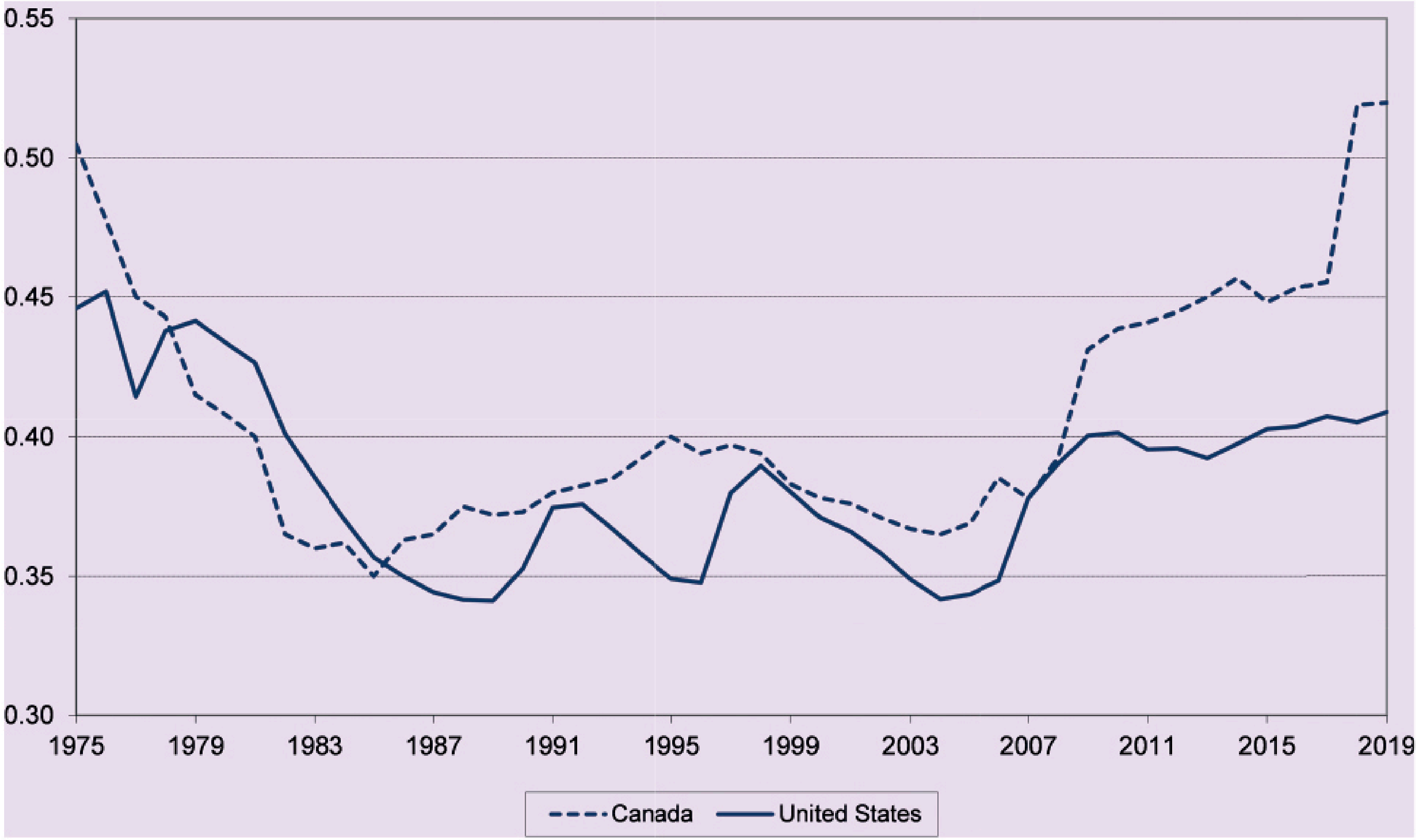


FIGURE 7.7 The Ratio of Minimum Wages to Average Wages, Canada and the United States, 1975–2019

Minimum Wages in Competitive Labour Market

- Suppose the labour market is competitive
- The market demand is downward sloping
- The market supply is upward sloping
- Minimum wages are introduced as a wage floor
  - Above the competitive equilibrium wage
- Creates excess supply of labour
  - Leads to unemployment

Minimum Wages in Competitive Labour Market

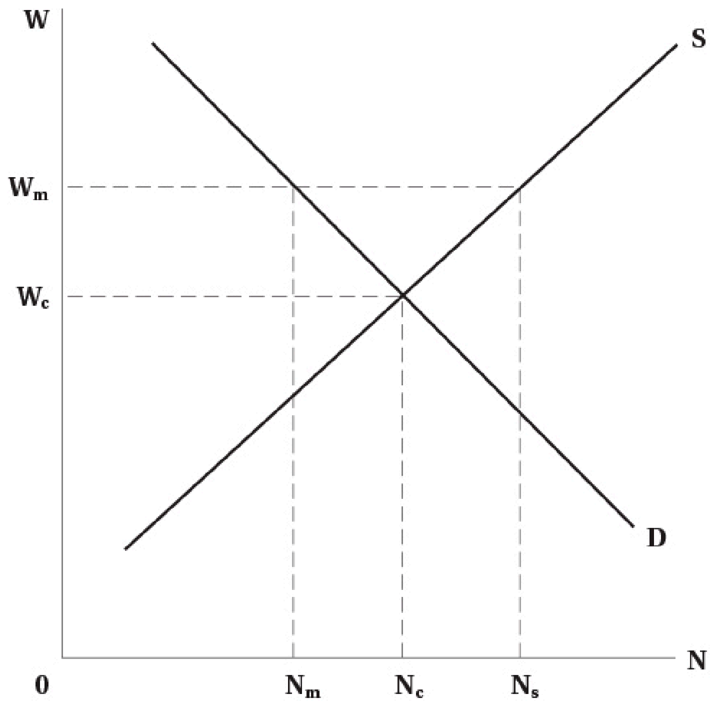


FIGURE 7.8 Minimum Wage

- Impact of minimum wage illustrated to the left
- Minimum wage is W_m above competitive level W_c
- Creates excess supply of labour
- Leads to unemployment $N_s - N_m$
  - People who would have been employed at competitive equilibrium $N_c - N_m$
  - Others drawn into market by higher wage $N_s - N_c$
-

Minimum Wages in Competitive Labour Market

- Employment effects depend on elasticity of demand
  - More elastic demand means larger employment effects
- In real world, industries with minimum wage tend to have elastic demands
  - Lots of substitutes for low-wage labour
  - Labour costs are a large share of total costs
- Nevertheless, we will see that minimum wages may not affect employment much in practice

Minimum Wages in Monopsony

- Effects of minimum wages are different in monopsonistic labour markets
- It is possible that minimum wages increase both wages and employment
  - Depending on where the minimum wage is set
- Graph on next slide illustrates this

Minimum Wages in Monopsony

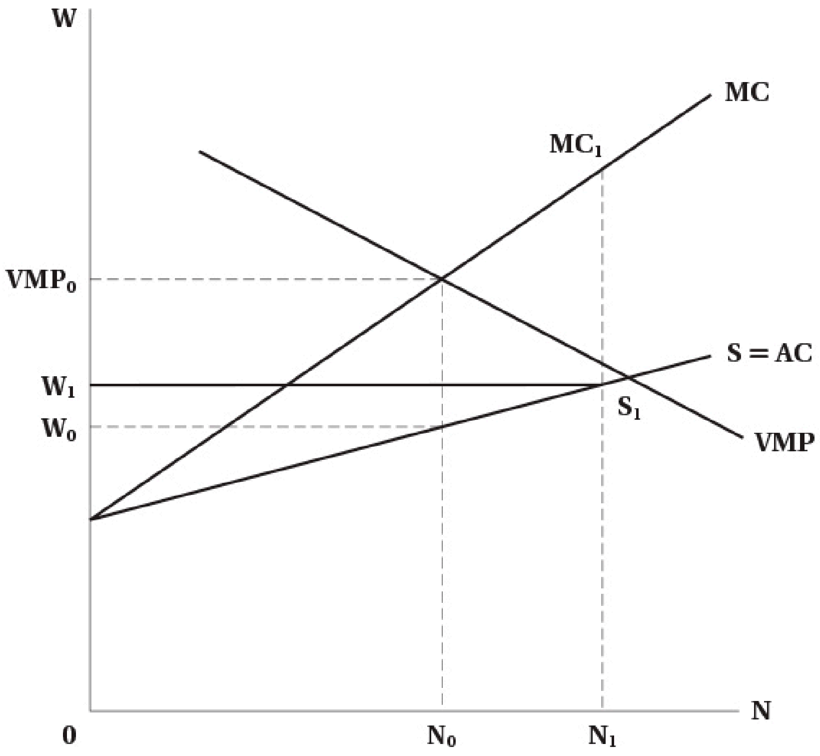


FIGURE 7.9 Monopsony and a Minimum Wage

- Graph shows standard monopsony
- Prior to minimum wage, wage is W_0 and employment is N_0
- Imagine minimum wage set above W_0 but below competitive wage
- MC curve is minimum wage up to S_1
  - Then jumps up to previous MC curve
- Optimize where MC = VMP
  - Employment increases to N_1

Minimum Wages in Monopsony

- Why would minimum wages cause employment to increase?
  - Previously, firm had to increase wages to attract more labour
  - With minimum wage, it does not
    - Everyone is paid the same wage
    - Wage increases no longer inhibit employment growth
  - Effectively, minimum wage lowers marginal cost
- This may be partly why we do not see large employment effects of minimum wages
- Note that monopsonists do not like minimum wages
  - They reduce their market power
  - Increases costs, reduces profits

Offsetting Factors for Minimum Wage

- As usual, in models we hold all else equal
- But in reality, many things are changing at once
  - Minimum wage could be accompanied by changes in demand
    - For example, fiscal stimulus increasing demand for goods and labour
  - Minimum wage could lead to changes in technology
    - May change demand for labour
  - Firms might choose to absorb minimum wage costs

Evidence

- Minimum wages are one of the most researched areas in labour economics
- Large amounts of evidence goes back at least 50 years
- Early studies looked at time trends in employment and minimum wages
  - Subject to a lot of bias
  - Things often trend together over time without direct causation
- Research in early 1990s used better methods
  - Natural experiments
  - Difference-in-differences

Evidence

- Most prominent study is Card and Krueger (1994)
  - Looked at fast food restaurants in New Jersey and Pennsylvania
  - New Jersey increased minimum wage, Pennsylvania did not
  - Used difference-in-differences to estimate employment effects
  - Found no negative employment effects from minimum wage increase
- Has been reevaluated several times
  - Conclusion remains that employment effects are small or non-existent

Evidence

- In Canada, minimum wages increase youth unemployment
    - Studies looked at minimum wage changes across provinces in 1990s
- Some Canadian studies look deeper at labour market dynamics
    - In any month, there is a lot of turnover in the labour market
    - Each month roughly 6% of employed people separate, 8% of non-employed people hired
    - Evidence shows minimum wages reduce both hiring and separations
    - Leads to no effect overall

Evidence

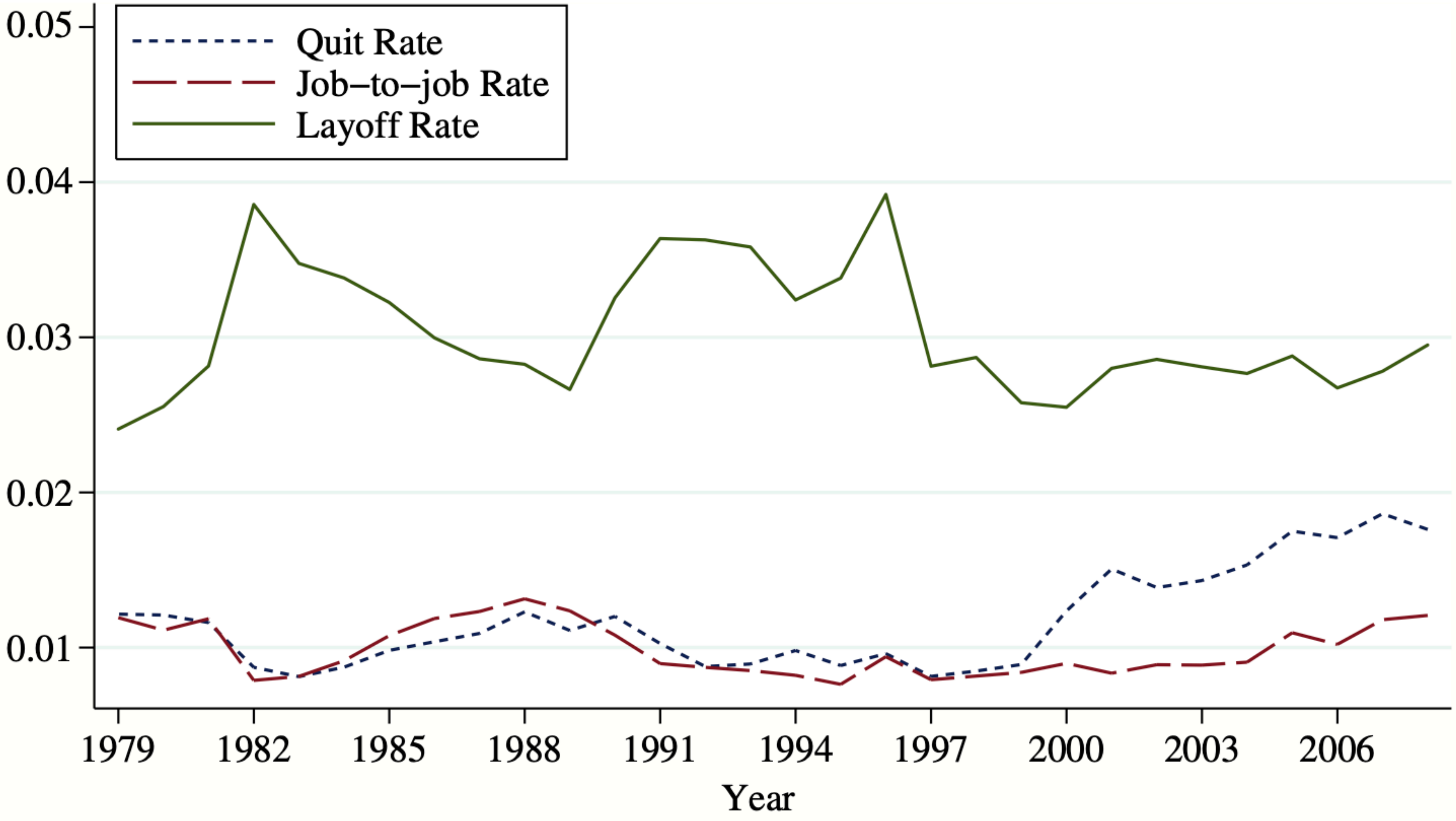


Fig. 4. *Layoff and Quit Rates, Low Skilled,* 1979–2008

Evidence

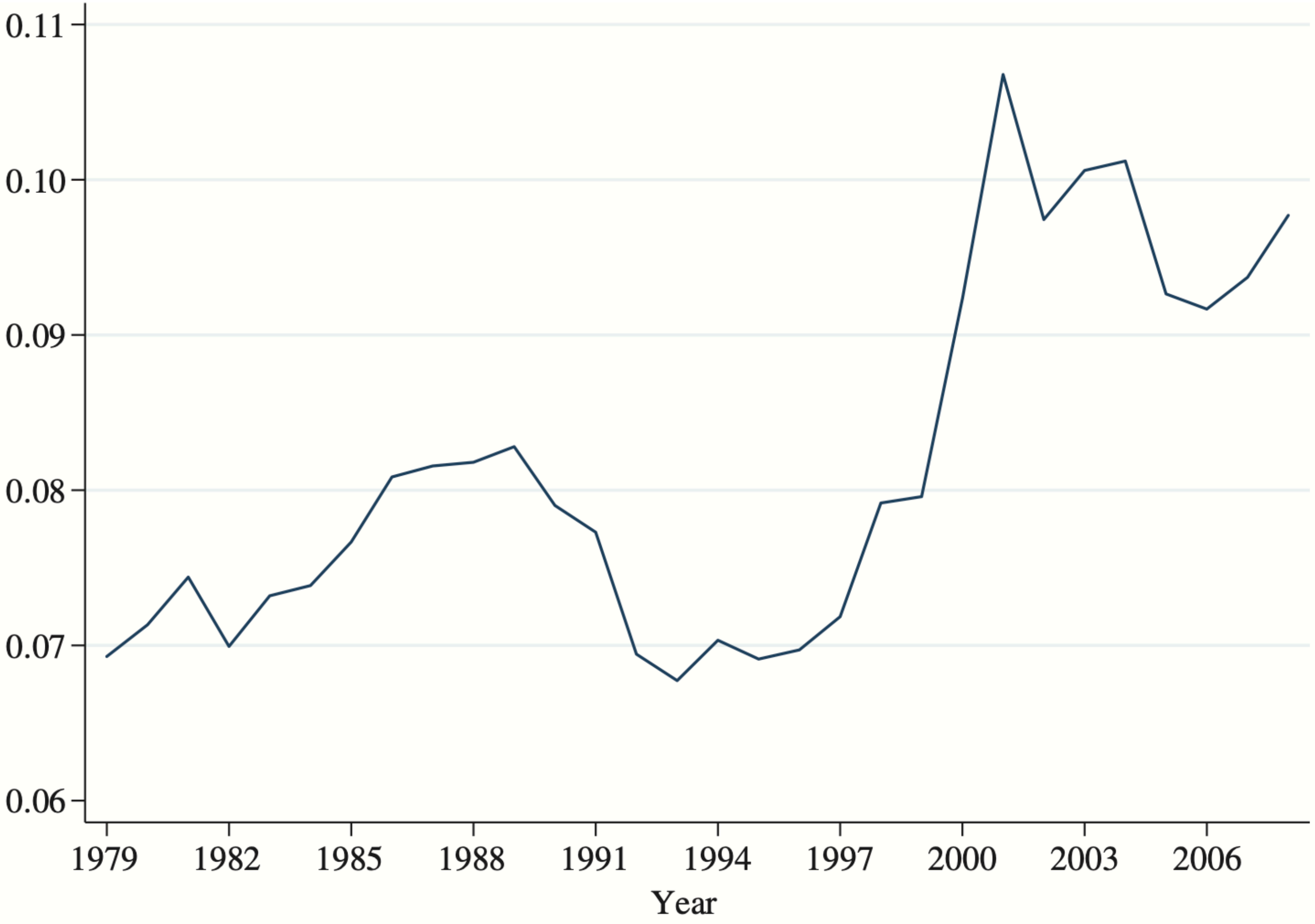


Fig. 5. *Hiring Rates, Low Skilled,* 1979–2008

Evidence

- Also recall that one point of miminum wage is to reduce poverty
- Evidence shows they are generally not successful
  - Both Canada and US studies show no link
- Why do they not work?
  - Many poor individuals do not work
  - Many people on minimum wages are not poor
    - Youth living in high income families

Summary

Summary

- We analyzed how wages and employment are determined in a single labour market
- Market power in the goods market reduces both wages and employment
- Market power in the labour market reduces wages and employment
- Minimum wages may reduce employment in competitive markets
- Minimum wages may increase employment in monopsonistic markets
- Empirical evidence shows minimum wages have small effects on employment